Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element K Initial Template - AT - 9 points

Element K: Reflection on the Design Project

Engineering Design Process Summary *(K1, K2)*

Summarize each step of the design process, including:

- Defining the problem (elements A-C) What went well? What didn't go well? Why?

What went well: We clearly identified our goal, to create a watering system for plants that is efficient, stable, and safe. We defined the design requirements and used early research to support our problem definition. **What didn't go well:** Our initial problem definition didn't fully consider chemical safety or maintenance access. These issues were only discovered later through expert feedback. **Why:** We focused heavily on the function and structure but not enough on the environmental and practical use factors at the start.

- Ideation (element D) What went well? What didn't go well? Why?

What went well: We generated several creative ideas and design sketches that helped guide our prototype. Collaboration and brainstorming helped us see different options for water distribution. **What didn't go well:** Some ideas were not realistic due to lack of materials or time. We didn't prioritize ideas early on, so we lost time deciding which design to move forward with. **Why:** We were more focused on creativity than feasibility at first and didn't check our ideas against real-world limits early enough.

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- STEM and Justification of design (elements E-F) What went well? What didn't go well? Why?

What went well: We used science to justify plant spacing and the misting system. Expert feedback helped confirm that our design choices made sense from both biological and structural standpoints. **What didn't go well:** We missed some research on chemical effects of the mold-resistant coating early on. **Why:** We were focused on structural performance and missed the chemical safety aspect in our initial research.

- Prototype building (element G): What went well? What didn't go well? Why?

What went well: The basic structure came together well, and we were able to build the prototype according to our blueprint. We used the correct tools and methods efficiently. **What didn't go well:** We had to make adjustments to the pipe stability and chemical safety cover after the first round of feedback. **Why:** Our first design was too basic and didn't anticipate issues like bending pipes or chemical exposure. We learned these through hands-on experience and testing.

- Testing (elements H-I) What went well? What didn't go well? Why?

What went well: We were able to test water flow and check for leaks and stability. These tests gave us data that helped guide improvements. **What didn't go well:** We didn't have advanced tools like moisture sensors to test how evenly the water was distributed or how much the plants absorbed. **Why:** Our testing tools were limited, which made it harder to collect objective data. Also, some tests were rushed due to time.

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- Evaluation (element J) What went well? What didn't go well? Why?

What went well: We received helpful and detailed feedback from both experts and stakeholders. Their input led to real improvements in our second prototype. **What didn't go well:** We didn't gather enough stakeholder feedback on the environmental and safety impact at first. **Why:** We focused more on structure and basic function, and less on the user and environmental safety side during the initial evaluation phase.

Lessons Learned About the Engineering Design Process *(K3)*

One major lesson we learned is that the design process is not just about building, it's about *iterating*. Feedback from users and experts is just as important as the initial design idea. Future designers should make time for early testing and be open to making changes, even late in the process. It's also important to think about how every part of the system interacts; structure, function, safety, and user experience. Planning for realistic testing tools and consulting with experts early can prevent bigger issues later. Overall, the engineering design process works best when it's flexible, detailed, and includes lots of feedback and testing at every step.